

Beyond Static Leaderboards: A Capability-Focused Extension for Live Evaluation of Time Series Foundation Models: Supplementary Material [Experiment, Analysis & Benchmark]

Supplementary material overview. This document expands the main paper in five parts. Appendix A specifies the benchmark-task, generation, verification, evaluation, aggregation, and replay contracts. Appendix B gives a standalone mathematical construction for each capability dimension (CD1–CD8), including its fixed structural choices, level-specific coordinate, affected fields, and interpretation boundary. Appendix C records the compute environment, evaluation support, aggregation rules, regeneration analysis, auxiliary-input removal results, and single-realization fine-tuning details. Appendix D binds the reported claim families to experiment identities, compact analysis records, coverage, and completed validation operations. Appendix E provides reviewer-facing artifact and reproduction instructions. The formulas define the generated diagnostic probes; they do not identify model-internal mechanisms or make the five levels comparable across capability dimensions.

A Protocol Details

This appendix specifies the construction, verification, execution, metric, and replay contracts behind the main paper. Its scope is the benchmark-generic protocol; the GIFT-Eval study supplies one empirical application. A CaFE-usable GIFT-Eval subset covers the source tasks admitted under the CaFE contract and therefore represents a narrower population than the complete official evaluation population of GIFT-Eval. The appendix follows the Data Generator’s three logical stages: Stage 1 maps a benchmark task from the source benchmark to References, Stage 2 constructs candidate probe groups, and Stage 3 applies Probe Verification followed by Group Verification, which combines complete-group qualification with task-level contract verification before admitting an evaluation bundle. The Evaluator runs a fixed model configuration on this bundle and reports four separate metrics. Adapter-specific split rules form part of the corresponding task and provenance contract. The eight capability dimensions are denoted CD1–CD8; their empirical results appear in Section C.

A.1 Stage 1: Preparation

A.1.1 Benchmark-Task Contract. Let t denote one benchmark task. Its adapter contract identifies the source data and fixes the forecast targets and their order, forecast-origin enumeration and horizon, benchmark-native input history, task-permitted auxiliary inputs and their visibility roles, observation policy, and validity conditions. A univariate setting has one prediction target, whereas a multivariate setting preserves multiple ordered targets as one forecasting object. Either target setting is covariate-informed when the task additionally provides task-permitted non-target covariates. The adapter maps the native representation to CaFE’s common interface while preserving these semantics; any model-required

channel decomposition is recorded as an inference adaptation rather than a change to the task’s target setting.

For the i th valid forecast origin of task t , the adapter returns a *Reference*, an adapter-qualified, source-bound forecast instance in CaFE’s common representation, written as

$$R_i = (H_i, Y_i^+, Z_i, M_i, o_i, t_i). \quad (1)$$

Here, H_i is the target history, Y_i^+ is the forecast truth, Z_i contains the auxiliary inputs permitted by t , M_i records the applicable observation masks, including the future-target observation mask that identifies scoreable cells, o_i is the forecast origin, and t_i is the source identifier. The provenance record resolves this identifier to the adapter, benchmark task, source record, and native window identity needed for reconstruction. The References of task t form

$$\mathcal{A}_t = \{R_i : R_i \text{ satisfies the adapter contract for } t\}. \quad (2)$$

Reference materialization. The task adapter owns forecast-origin enumeration and future-label eligibility. It may require a fully observed horizon or retain a future-target observation mask, with the resulting scoreability state carried by the Reference.

Information boundary. History repair, when required, follows a deterministic adapter-resolved rule derived from history. The corresponding scale policy declares whether repaired values or originally observed pairs enter history-based statistics. The source history, task contract, and recorded augmentation seed determine dimension selection and every structural, level, and random choice; future target values enter the workflow as scoreable truth after these choices. The future-target observation mask may constrain where a constructed effect must remain visible and determines which future cells are scoreable. Horizon visibility follows the declaration of task t : known-future dynamic covariates extend across the horizon, and past-only fields end at the forecast origin.

Stage output and provenance. Stage 1 records the adapter, task, source, and native-window identities together with the bound task contract for each R_i . These records support later contract checks and source reconstruction under the verification policy selected for the evaluation round.

A.2 Stage 2: Generation

A.2.1 Five-Level Probe-Group Contract. For Reference R_i and capability dimension c , Stage 2 selects a shared structural configuration ξ_{ic} . It fixes the affected targets and the dimension-specific temporal form or predictive relation across all five levels. The index $\ell \in \{1, \dots, 5\}$ marks an ordered position within c , and $\lambda_{ic\ell}$ is its dimension-specific parameter. The corresponding diagnostic probe and candidate probe group are

$$\begin{aligned} V_{ic\ell} &= T_c(R_i; \xi_{ic}, \lambda_{ic\ell}), \\ \tilde{\xi}_{ic} &= (R_i, V_{ic1}, \dots, V_{ic5}), \end{aligned} \quad (3)$$

where T_c is the dimension-specific transformation. Thus, a candidate probe group contains one Reference and exactly five diagnostic probes. Group members preserve the task, source identifier, origin, horizon, target order, visibility roles, and shared configuration ξ_{ic} . The level parameter $\lambda_{ic\ell}$ is the only ordered control, although applying it may determine dimension-specific level details such as the compatible phase used by CD5. Because this parameter may represent amplitude, change location, period count, or event spacing, the level order and scale are local to each dimension. The Reference serves as the baseline, and diagnostic-probe level indexing begins at one.

Field-level transformation contract. The dimension-specific transformation can be written componentwise as

$$V_{ic\ell} = (H_i + \Delta H_{ic\ell}, Y_i^+ + \Delta Y_{ic\ell}, Z_i + \Delta Z_{ic\ell}, M_i, o_i, t_i). \quad (4)$$

The deltas identify the fields changed by dimension c ; zero deltas retain the corresponding Reference fields. CD1–CD7 act on target histories and their matched diagnostic futures, so $\Delta Z_{ic\ell} = 0$. CD8 adds a task-permitted dynamic-covariate impulse and its matched target response, with its future covariate component restricted by the task’s visibility declaration. Thus, every transformation changes only the dimension-declared fields of R_i .

Deterministic generation-seed variation. Level comparisons keep all generation-seed-dependent choices represented by ξ_{ic} matched. Randomness is domain-separated by object role: group-level structure draws are keyed by the source-instance identifier, capability-dimension identifier, a structure tag, and the recorded augmentation seed, without the level; level-specific draws additionally include ℓ . CD7 and CD8 first enumerate bounded deterministic seed-independent qualification pools, after which the recorded augmentation seed selects one qualified structure. The generated contract records the selected targets, shared structure, level parameter, seed identity, and sufficient state to replay T_c . The future-target observation mask may constrain required future-effect support, but future target values and model outputs do not enter structure selection. Construction completes before model inference.

Structure selection and level feasibility. Each mechanism first determines applicability and constructs its permitted structure. Cross-Series Dependence and Covariate Response additionally form bounded, history-qualified candidate sets before seed selection: the recorded augmentation seed chooses one structure from the qualified set, and that structure is then shared by L1–L5. For an amplitude-controlled dimension, the generator normalizes the selected raw component by its full-history macro source distance and intersects each nominal level interval with the analytically feasible range induced by the required model-visible contexts, channel bounds, and any applicable future-effect requirement. If raw component g_{ic} has full-history macro distance q_{ic} , level ℓ applies physical gain $a_{ic\ell} = \lambda_{ic\ell}/q_{ic}$ and sets the corresponding delta to $a_{ic\ell}g_{ic}$. A structure qualifies when every level has a nonempty effective interval. The level-inclusive draw selects $\lambda_{ic\ell}$ within that interval; CD2, CD4, and CD5 instead retain period count, transition location, and event spacing as their dimension-specific level parameters.

Source-distance evidence. Qualification uses a common distance to keep each diagnostic change visible and bounded relative to its source. Let $\mathcal{J}_{ic}$ be the affected target set, C a protocol-declared history view, and $\sigma_{id} > 0$ the source-history standard deviation for target d . A nonfinite or degenerate standard deviation falls back first to the standard deviation of first differences and then to the declared positive value 1. This quantity normalizes construction distance, while the MASE scale s_{id} normalizes forecast error. Stage 2 records

$$d_{ic\ell C} = \frac{1}{|\mathcal{J}_{ic}|} \sum_{d \in \mathcal{J}_{ic}} \sqrt{\frac{1}{|C|} \sum_{u \in C} \left(\frac{H_{ic\ell,u,d}^V - H_{ic\ell,u,d}^R}{\sigma_{id}} \right)^2}. \quad (5)$$

This distance uses R_i as the comparator at every level. For every required model-visible context, the macro value must lie in $[0.10, 2.0]$; each affected channel has an upper bound of 3.0. Structural variables such as period count or event spacing serve as level parameters and express structural progression.

Full-history construction and context qualification. The transformation law and its level parameter are defined on the complete benchmark-native history admitted by task t . Full-history source distance calibrates amplitude-controlled levels, while qualification evaluates the same realized transformation on every distinct history suffix that a selected model configuration may receive. The Evaluator applies the frozen transformation first and then projects the resulting history to the model’s fixed context. Context projection therefore changes the visible prefix boundary while preserving the selected structure, level parameter, target roles, and forecast origin.

A.2.2 Generation Output. Stage 2 constructs each candidate probe group and records its Reference link, target identities, shared structural configuration, level parameters, per-probe evidence, and replay state. Stage 3 then evaluates each realized probe, qualifies the candidate as a complete five-level group, and applies the round’s task-level verification policy before assembling the evaluation bundle. Auxiliary-removal views are not generated probes and do not participate in Probe Verification or complete-group qualification. For CD6–CD8, the Evaluator derives an ablated-input view from an admitted diagnostic probe under the independent inference-ablation contract specified below. The eight constructions appear in Section B.

A.3 Stage 3: Verification

Stage 3 applies two verification steps to the candidate generation output. Probe Verification evaluates each realized diagnostic probe, and Group Verification treats the five levels as one matched unit before applying the evaluation round’s task-level policy. The resulting report is bound to the generation manifest; only an accepted report admits the corresponding bundle to inference.

A.3.1 Probe Verification. Probe Verification evaluates each realized diagnostic probe $V_{ic\ell}$ against three recorded requirements: consistency with the structure selected in Stage 2, source-distance bounds on every required model-visible history suffix, and any capability-specific horizon or future-effect support condition. The source-distance evidence uses Equation (5): every required history

view must have macro distance in $[0.10, 2.0]$, and every affected channel must have distance at most 3.0. The verification record preserves separate pass or non-numeric failure states for these checks rather than treating an unavailable or failed probe as a zero-valued result.

A.3.2 Group Verification.

Complete-group qualification. A candidate probe group $\tilde{G}_{ic}$ qualifies by satisfying all of the following conditions:

- (1) capability dimension c is applicable to R_i , all five levels are present, and every level passes Probe Verification;
- (2) the five levels share the Reference, fields, and structural configuration required by Equations (3) and (4); and
- (3) the ordered level parameters and other cross-level construction invariants of dimension c hold across the complete group.

Qualification applies to the complete five-level group. A candidate is retained as one unit after all five levels pass; a missing or failed level assigns rejected status to the entire group. Each (R_i, c) pair therefore contributes exactly five matched diagnostic probes or a recorded non-numeric state, while its Reference remains in $\mathcal{A}_t$. A passing group is denoted G_{ic} , and the qualified groups of task t form

$$\mathcal{P}_t = \{G_{ic} : \tilde{G}_{ic} \text{ passes group qualification}\}. \quad (6)$$

An inapplicable or rejected construction is recorded with its reason and a non-numeric state. A minimum future-change condition is a group-level construction invariant only for dimensions that explicitly declare one; for all other dimensions, truth-change sufficiency is evaluated later as an NRMSE scoreability condition and does not remove an otherwise scoreable MASE value.

Task-level contract verification. The verification report is bound to the generation manifest. The research policy scans every emitted treatment contract and checks its recorded source-distance, mechanism-scoring, and applicable horizon-support evidence. It records the contract counts, failed checks, and acceptance outcome.

For source adapters with an exact-replay policy, verification additionally checks manifest and artifact identities and reconstructs the contracts for References and diagnostic probes from source data. Exact replay is recorded as a stronger verification level rather than inferred from research-policy acceptance. Write $\text{VerOK}_t = 1$ when the declared policy returns an accepted report bound to the generation manifest.

Verified evaluation bundle. For an accepted task, Stage 3 assembles

$$\mathcal{D}_t = (\mathcal{A}_t, \mathcal{P}_t) \quad \text{if } \text{VerOK}_t = 1. \quad (7)$$

The bundle carries the links between each Reference and diagnostic probe together with the accepted verification policy and report. A failed report prevents the bundle from entering inference. Any CD6–CD8 ablated-input view is derived from an admitted probe after verification and recorded under a separate inference-ablation contract.

A.4 Evaluator

The Evaluator reports four separate result objects. Reference MASE and diagnostic-probe MASE quantify error on unchanged

References and constructed probes, respectively. Paired forecast-change NRMSE measures target-side response fidelity to the constructed capability intervention. For CD6–CD8, Input Sensitivity (AMASE) reports the assessed-target MASE change after direct removal of the designated auxiliary input.

A.4.1 Model Configuration and Execution States. Intervention-response comparisons use one fixed model-execution contract across paired objects. Model configuration m therefore binds the model to every choice that can alter its input or forecast, including model identity, access mode, context policy, preprocessing, target and covariate mappings, and permitted adaptations. It is shared by the Reference and all five diagnostic probes. In an eligible auxiliary-removal pair, the configuration and randomization policy are also fixed; only the declared input view changes.

Input projection and output restoration. The Evaluator applies one fixed execution plan to every Reference and diagnostic probe in an admitted evaluation bundle. It first checks horizon and auxiliary-input compatibility, then applies the model’s context projection to the already constructed history of the corresponding object. A *native* multivariate interface receives the target panel as one forecasting object. For the same multivariate task, an *adapted* independent-univariate interface produces channel-wise requests, restores the native target order before metric computation, and defines a distinct model configuration. When no permitted mapping satisfies the horizon, target, and input contract, the view contributes to the unsupported-coverage count and no request is issued. A supported request may complete or fail; malformed, shape-incompatible, or nonfinite responses are recorded as failed requests. A valid forecast matches the task horizon, target order, expected shape, and finite-value requirements. Auxiliary-removal execution is restricted to configurations that actually receive the input being removed.

Metric scoreability forms a third axis. Each metric constructs its numeric population from valid forecasts and its required pairings. NRMSE records scored, low-truth-effect, and no-observed-effect states, while the execution and analysis manifests retain the available coverage counts for the other outputs. Numeric aggregation uses only computed values.

A.4.2 Reference-History Scale and MASE. Mean absolute scaled error (MASE) [1] measures forecast error across target scales. Let L_i be the history length, let p_t be the adapter-resolved seasonality, and set $\tau_t = \min(\max(1, p_t), \max(1, L_i - 1))$. For target d , let $\mathcal{Q}_{id}(\tau_t)$ contain the history-index pairs admitted by the adapter-resolved scale policy. A complete repaired-history policy admits the repaired cells, whereas a history-observation-preserving scale policy admits pairs whose two source cells are observed. The Reference-history scale is

$$s_{id} = \frac{1}{|\mathcal{Q}_{id}(\tau_t)|} \sum_{u \in \mathcal{Q}_{id}(\tau_t)} |H_{i,u,d} - H_{i,u-\tau_t,d}|. \quad (8)$$

When this seasonal-pair scale is unavailable or at most 10^{-8} , the evaluator applies the same calculation to adjacent pairs admitted by the scale policy; a remaining unavailable or degenerate scale receives the declared positive fallback value 1. For each probe group,

the Reference, diagnostic probes, and any eligible full/ablated inference pair share the resulting s_{id} .

For object type $X \in \{R, V\}$, let Ω_i^X contain its observed future cells. Its instance-level score is

$$\text{MASE}_i^X = \frac{1}{|\Omega_i^X|} \sum_{(h,d) \in \Omega_i^X} \frac{|\hat{Y}_{i,h,d}^X - Y_{i,h,d}^X|}{s_{id}}. \quad (9)$$

Within a fixed task and result unit, the evaluator averages instance-level values equally. Lower MASE therefore denotes smaller forecast error relative to the Reference history’s variation scale. For $X = R$, this result object is Reference MASE; for $X = V$, it is diagnostic-probe MASE. The input-ablation comparison uses the same formula restricted to the assessed target under the full and ablated input views.

A.4.3 NRMSE for Intervention-Response Fidelity. Fix task t , model configuration m , capability dimension c , and level ℓ . For an affected and observed future cell $k = (h, d)$ of instance i , define the truth and forecast changes between the diagnostic probe and its Reference as

$$\Delta Y_{ik} = Y_{ik}^V - Y_{ik}^R, \quad \Delta \hat{Y}_{ik} = \hat{Y}_{ik}^V - \hat{Y}_{ik}^R. \quad (10)$$

Let Ω_i^Δ contain the affected observed cells declared by T_c , and use the Reference-history scale $s_{ik} = s_{id}$. The standardized truth-change RMS is

$$q_i = \sqrt{\frac{1}{|\Omega_i^\Delta|} \sum_{k \in \Omega_i^\Delta} \left(\frac{\Delta Y_{ik}}{s_{ik}} \right)^2}. \quad (11)$$

The eligible set $\mathcal{J}_{t,m,c,\ell}^\Delta$ contains instances with two valid forecasts, nonempty Ω_i^Δ , and $q_i \geq 0.05$. The task-level normalized root-mean-square error (NRMSE) is

$$\text{NRMSE}_{t,m,c,\ell} = \sqrt{\frac{\sum_{i \in \mathcal{J}_{t,m,c,\ell}^\Delta} \sum_{k \in \Omega_i^\Delta} \left(\frac{\Delta \hat{Y}_{ik} - \Delta Y_{ik}}{s_{ik}} \right)^2}{\sum_{i \in \mathcal{J}_{t,m,c,\ell}^\Delta} \sum_{k \in \Omega_i^\Delta} \left(\frac{\Delta Y_{ik}}{s_{ik}} \right)^2}}. \quad (12)$$

The ratio pools standardized squared sums over eligible instances and cells and requires a nonempty set with positive truth-change energy. Instances below the 0.05 truth-change threshold remain available to MASE and retain a non-numeric NRMSE state. NRMSE is zero for exact agreement between forecast and truth changes and one when forecast changes equal zero over the included cells. It is the paired forecast-change response score for the complete capability intervention encoded by T_c .

A.4.4 Input Sensitivity via Direct Auxiliary-Input Removal. This inference ablation is defined only for model/view pairs that actually expose the designated auxiliary input. For CD6, the highest-absolute-loading target is assessed and the ablated view retains only that target, deleting every other target channel. For CD7, the responder is assessed and the ablated view retains only the responder, again deleting every other target channel. For CD8, the target panel is retained in both views while the ablated view deletes

the designated dynamic-covariate carrier together with its corresponding name, availability, future-visibility, and type metadata. Independent-univariate configurations that never receive other target channels are excluded from the CD6/CD7 comparison, and configurations that do not receive the CD8 carrier are excluded there.

Let Ω_i^{abl} contain the observed future cells of the dimension-specific assessed target. Let A_i^V and A_i^{abl} denote Equation (9) restricted to Ω_i^{abl} for the full diagnostic-probe and ablated-input views, respectively. The two views share the assessed-target history and future, forecast origin, horizon, observation mask, and Reference-history scale. Let $\mathcal{J}_{t,m,c,\ell}^{\text{abl}}$ contain the eligible pairs for which both forecasts are scoreable. Input Sensitivity is

$$\Delta \text{MASE}_{t,m,c,\ell} = \frac{1}{|\mathcal{J}_{t,m,c,\ell}^{\text{abl}}|} \sum_{i \in \mathcal{J}_{t,m,c,\ell}^{\text{abl}}} (A_i^{\text{abl}} - A_i^V). \quad (13)$$

Positive values mean that auxiliary-input removal increases assessed-target error, zero denotes no average signed difference, and negative values mean that removal decreases the error. The quantity records net predictive contribution under the declared deletion and does not by itself identify a causal or internal model mechanism.

A.4.5 Recorded Analysis Aggregation and Uncertainty. The current recorded analyses use the following CAFE-native task-equal contract; this contract does not establish source-benchmark evaluator or leaderboard parity. For a result unit r selected by the analysis contract for suite-level reporting, let $\text{Metric}_{t,r,m}$ be the task-level value for model configuration m , and let $\mathcal{T}_{r,m}$ be the tasks on which that value is computed. The unit identifies one metric and its evaluation object or comparison condition, plus a capability dimension and level where applicable. Its recorded task-equal suite result is

$$\text{SuiteMetric}_{r,m} = \frac{1}{|\mathcal{T}_{r,m}|} \sum_{t \in \mathcal{T}_{r,m}} \text{Metric}_{t,r,m}, \quad |\mathcal{T}_{r,m}| > 0. \quad (14)$$

Thus, each benchmark task contributes one value regardless of its number of References or qualified groups. The suite row records the task-set cardinality; the contributing task identities can be reconstructed from the bound task-level upstream artifacts.

For example, suppose task A has two scoreable instances with values 1 and 3, while task B has eight scoreable instances with value 10. The task-level values are 2 and 10, so Equation (14) gives $(2 + 10)/2 = 6$. The row-level pooled estimand gives $(1 + 3 + 8 \times 10)/10 = 8.4$ and assigns task B four times the weight of task A. The recorded analyses use this task-equal estimand.

Within result unit r , the comparison task set for model configurations m_1 and m_2 is $\mathcal{T}_{r,m_1} \cap \mathcal{T}_{r,m_2}$. The recorded analysis computes one paired difference per common task and averages them with equal task weight. The bootstrap resamples tasks for a single-model result and paired task differences for a model comparison; its seed, repetition count, interval rule, and realized task identities remain in the analysis contract. Reference MASE, diagnostic-probe MASE, paired forecast-change NRMSE, and Input Sensitivity remain separate result objects with their own directions, domains, and scoreable support.

A.5 Provenance, Integrity, and Replay

To keep materializations comparable, the Data Generator’s three logical stages produce immutable contracts across the execution phases of generation, verification, inference, and analysis. Experiments with different protocol or identity fields remain separate analysis populations.

Identity records. The experiment record binds the benchmark adapter and its configuration, source identity, task identities and the native window identity of each Reference, capability-dimension roster, generation configuration and seed, verification mode, and model configurations. Each stage contract records the identities and relations of its inputs and outputs. These fields keep the following provenance chain explicit:

$$\mathfrak{C}_t \xrightarrow{\text{Stages 1-2}} \mathfrak{G}_t \xrightarrow{\text{Stage 3}} (\mathcal{D}_t, \mathfrak{B}_t) \xrightarrow{m} \mathfrak{P}_{t,m} \xrightarrow{\text{Metrics}} \mathfrak{A}_{t,m}, \quad (15)$$

where $\mathfrak{C}_t$ denotes the bound benchmark-task input, $\mathfrak{G}_t$ its compact generation contracts, $\mathfrak{B}_t$ the verification report, $\mathfrak{P}_{t,m}$ the predictions, and $\mathfrak{A}_{t,m}$ the metric artifacts. The analysis binds the task-level upstream artifacts and records each result unit’s task-set cardinality; contributing identities can be reconstructed from those upstream artifacts.

Verification and replay levels. Research contract verification checks the recorded treatment evidence covered by its declared policy. Exact source replay additionally verifies source and artifact identities and reconstructs the contracts for References and diagnostic probes when the adapter supports that policy. Inference-and-analysis replay also uses the bound model and analysis configurations, including any separately declared input-ablation views. Each report records the highest completed operation and its outcome.

Integrity verification compares each recorded object with its bound identity and content. Replay status additionally requires the corresponding reconstruction and comparison. Source-to-result reproduction requires the complete recorded chain and declared comparison rule, while independent reproduction also requires the pre-declared independent unit and non-overlapping identities. These remain distinct evidence levels.

B Mathematical Constructions of the Capability Dimensions

This appendix gives the mathematical construction used for each of the eight capability dimensions. The formulas parameterize the implemented construction families at the level needed to recover which fields change, which structural choices remain fixed across L1–L5, and which coordinate varies with the level. They are construction definitions rather than claims that a generated component is perfectly isolated or that an observed forecast response identifies a model-internal mechanism.

B.1 Shared Notation and Level Contract

For Reference R_i , let L_i and H_i^{hor} denote its history length and forecast horizon. We use $u \in \{0, \dots, L_i + H_i^{\text{hor}} - 1\}$ on the concatenated

target path

$$X_{i,d}(u) = \begin{cases} H_i[u, d], & 0 \leq u < L_i, \\ Y_i^+[u - L_i, d], & L_i \leq u < L_i + H_i^{\text{hor}}, \end{cases} \quad (16)$$

where d indexes prediction targets. Let $U_{i,q}(u)$ denote task-permitted dynamic covariate q on its declared visible range. A diagnostic probe changes these fields by

$$X_{ict}^V = X_i + \Delta X_{ict}, \quad U_{ict}^V = U_i + \Delta U_{ict}. \quad (17)$$

The future values in Equation (16) provide scoreable truth after construction; they do not enter the selection of the structural configuration ξ_{ic} .

For the amplitude-controlled dimensions CD1, CD3, CD6, CD7, and CD8, let $G_{ic}^X(u, d)$ and $G_{ic}^U(u, q)$ be the raw target and covariate components fixed by ξ_{ic} . With affected-target set $\mathcal{J}_{ic}$ and source-history standard-deviation scale σ_{id} , define the full-history macro normalized RMS

$$\mathcal{D}_i(G_{ic}^X; \mathcal{J}_{ic}) = \frac{1}{|\mathcal{J}_{ic}|} \sum_{d \in \mathcal{J}_{ic}} \sqrt{\frac{1}{L_i} \sum_{u=0}^{L_i-1} \left(\frac{G_{ic}^X(u, d)}{\sigma_{id}} \right)^2}. \quad (18)$$

Writing $q_{ic} = \mathcal{D}_i(G_{ic}^X; \mathcal{J}_{ic})$, the level- ℓ construction is

$$(\Delta X_{ict}, \Delta U_{ict}) = \frac{\lambda_{ic\ell}}{q_{ic}} (G_{ic}^X, G_{ic}^U). \quad (19)$$

The five nominal intervals for $\lambda_{ic\ell}$ are $[0.10, 0.15]$, $[0.17, 0.22]$, $[0.25, 0.32]$, $[0.36, 0.46]$, and $[0.50, 0.65]$, intersected with the feasible interval of the selected structure. CD2, CD4, and CD5 instead use period count, transition location, and event gap as their ordered coordinates. Thus, L1–L5 are dimension-local coordinates rather than a universal difficulty scale.

Figure 1 illustrates the constructed components defined below.

B.2 CD1: Trend

CD1 selects a nonempty target subset $\mathcal{J}_{i1}$ with history-stable trend signs $s_{id}^{\text{tr}} \in \{-1, 1\}$. For each selected target, the raw component is

$$G_{i1}^X(u, d) = \mathbf{1}\{d \in \mathcal{J}_{i1}\} \sigma_{id} s_{id}^{\text{tr}} p_{i1}(u), \quad \Delta X_{i1\ell} = \frac{\lambda_{i1\ell}}{q_{i1}} G_{i1}^X, \quad (20)$$

where p_{i1} is one structure shared by all levels:

$$\begin{aligned} p_{\text{lin}}(u) &= \frac{u}{L_i - 1}, \\ p_{\text{del}}(u) &= \frac{(u - u_0)_+}{L_i - 1}, \\ p_{\text{curv}}(u) &= \left(\frac{u}{L_i - 1} \right)^r, \\ p_{\text{pw}}(u) &= \frac{1}{L_i - 1} \sum_{v=0}^{u-1} b_{\lfloor (v+\phi)/K \rfloor \bmod B}, \\ p_{i1} &\in \{p_{\text{lin}}, p_{\text{del}}, p_{\text{curv}}, p_{\text{pw}}\}. \end{aligned} \quad (21)$$

The four alternatives are whole-history linear, delayed linear, slow-curvature, and recurring piecewise-linear continuation. The onset u_0 , curvature r , knot interval K , phase ϕ , slope cycle $(b_0, \dots, b_{B-1})$,

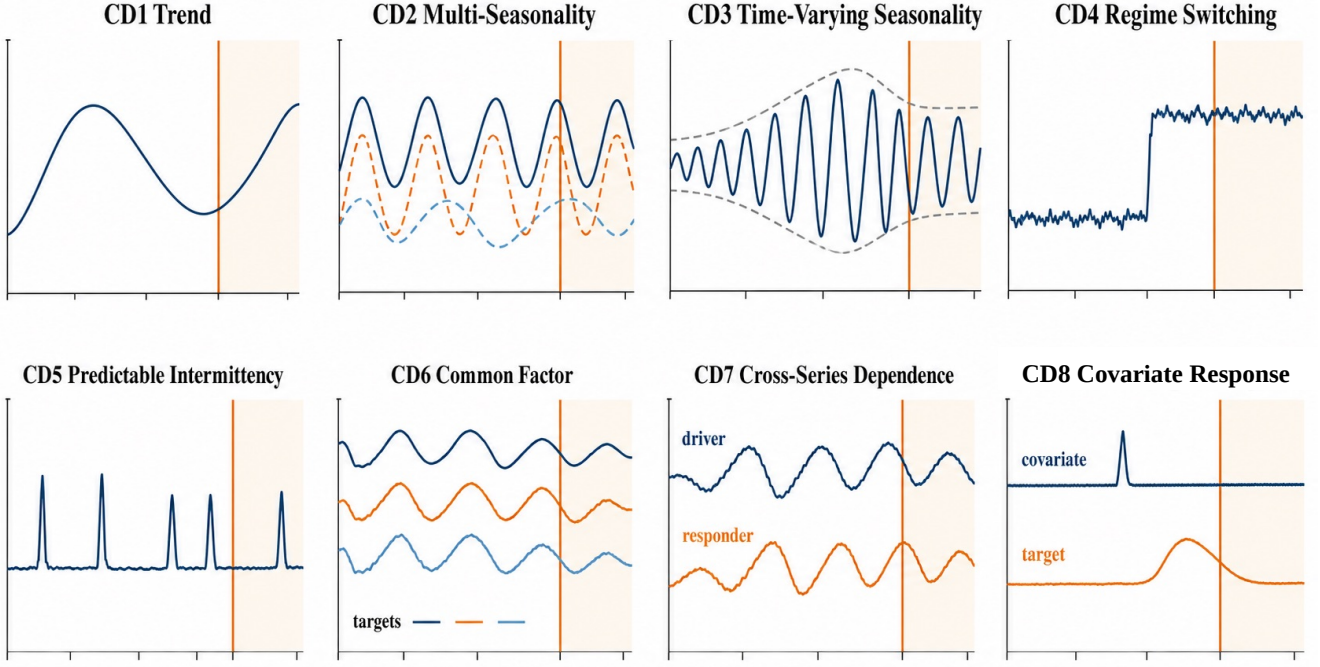


Figure 1: Illustrative controlled components for CD1–CD8. Each panel joins the modified history to the analytic continuation over the forecast horizon while retaining the benchmark task’s target and auxiliary-input roles.

affected targets, and signs belong to ξ_{i1} ; only normalized magnitude $\lambda_{i1\ell}$ changes across levels. Construction requires a stable history-derived direction and continues the same profile law through the horizon.

B.3 CD2: Multi-Seasonality

For target d , let the selected anchor and five additional independent periods be $P_{id0}, \dots, P_{id5}$. Component r is either a sinusoid or a two-harmonic waveform,

$$h_{idr}(u) = \sum_{m=1}^{M_{idr}} a_{idrm} \sin\left(\frac{2\pi mu}{P_{idr}}\right) + \sum_{m=1}^{M_{idr}} b_{idrm} \cos\left(\frac{2\pi mu}{P_{idr}}\right), \quad (22)$$

$$M_{idr} \in \{1, 2\}.$$

Each component is history-normalized to the target scale before aggregation. Level $\ell \in \{1, \dots, 5\}$ contains the anchor and the first ℓ additional periods:

$$S_{i2\ell}(u, d) = \sum_{r=0}^{\ell} h_{idr}(u), \quad \Delta X_{i2\ell}(u, d) = \frac{\rho_{i2}}{\mathcal{D}_i(S_{i2\ell}; \mathcal{J}_{i2})} S_{i2\ell}(u, d), \quad (23)$$

where $\mathcal{J}_{i2}$ contains all targets and the shared full-history distance $\rho_{i2} \in [0.22, 0.28]$ is fixed across the five levels. Hence the level coordinate is the number of additional independent periods, producing

two through six periods while holding aggregate history amplitude fixed. Periods, phases, waveform families, and their order are selected from history-qualified or protocol-generated candidates without future target values.

B.4 CD3: Time-Varying Seasonality

For an eligible target d , the selected carrier, its quadrature component, and a slower envelope are

$$\begin{aligned} c_{id}(u) &= a_{id}^c \sin(\omega_{id}^c u) + b_{id}^c \cos(\omega_{id}^c u), \\ c_{id}^\perp(u) &= b_{id}^c \sin(\omega_{id}^c u) - a_{id}^c \cos(\omega_{id}^c u), \\ e_{id}(u) &= a_{id}^e \sin(\omega_{id}^e u) + b_{id}^e \cos(\omega_{id}^e u), \quad \omega_{id}^e < \omega_{id}^c. \end{aligned} \quad (24)$$

The raw component uses the carrier for amplitude modulation and the quadrature carrier for the periodic phase-drift component:

$$\begin{aligned} b_{id}^{\text{mod}}(u) &= \begin{cases} c_{id}(u), & \text{amplitude modulation,} \\ c_{id}^\perp(u), & \text{periodic phase drift,} \end{cases} \\ G_{i3}^X(u, d) &= \mathbf{1}\{d \in \mathcal{J}_{i3}\} e_{id}(u) b_{id}^{\text{mod}}(u), \\ \Delta X_{i3\ell} &= \frac{\lambda_{i3\ell}}{q_{i3}} G_{i3}^X. \end{aligned} \quad (25)$$

History fits determine the carrier and slower envelope; the seed selects from a qualified leading pool and fixes the nonempty target subset and modulation type. Level ℓ changes only the normalized magnitude. Qualification requires the envelope to remain active on the declared fraction of observed horizon positions.

B.5 CD4: Regime Switching

Let $\tau_{i4\ell}$ be the level-specific transition location, w_{i4} the shared transition width, and

$$p(u) = \text{clip}\left(\frac{u - \tau}{w}, 0, 1\right),$$

$$r_\kappa(u; \tau, w) = \begin{cases} \mathbf{1}\{u \geq \tau\}, & \kappa = \text{step}, \\ p(u), & \kappa = \text{ramp}, \\ 3p(u)^2 - 2p(u)^3, & \kappa = \text{sigmoid}, \end{cases} \quad (26)$$

With affected-target signs s_{id}^{rg} , the five treatments are

$$\Delta X_{i4\ell}(u, d) = \alpha_{i4} \mathbf{1}\{d \in \mathcal{J}_{i4}\} \sigma_{id} s_{id}^{\text{rg}} r_{\kappa_{i4}}(u; \tau_{i4\ell}, w_{i4}). \quad (27)$$

The seed fixes $\mathcal{J}_{i4}$, the signs, transition family κ_{i4} , and width. The ordered coordinate is $\tau_{i4\ell}/L_i$; higher levels place the same state change closer to the forecast origin and therefore expose less post-transition history. The shared gain α_{i4} is chosen once for the five-level group to meet the history-distance contract.

B.6 CD5: Predictable Intermittency

Let w_{i5} be the shared pulse width. For offset r , the selected pulse kernel is

$$\kappa_{i5}(r) = \begin{cases} 1, & 0 \leq r < w_{i5}, \quad \text{rectangular}, \\ 1 - \frac{|r - (w_{i5} - 1)/2|}{\max(1, (w_{i5} + 1)/2)}, & 0 \leq r < w_{i5}, \quad \text{triangular}, \\ \exp\left[-\frac{(\log 2)r}{\max(1, w_{i5}/2)}\right], & 0 \leq r < w_{i5}, \quad \text{exponential}, \\ 0, & \text{otherwise.} \end{cases} \quad (28)$$

For level ℓ , event centers $\mathcal{E}_{i5\ell}$ follow mean gap $g_{i5\ell}$ with bounded alternating jitter $\epsilon_{i5\ell}$, so consecutive gaps are $\max(w_{i5} + 1, g_{i5\ell} \pm \epsilon_{i5\ell})$. The target increment is

$$\Delta X_{i5\ell}(u, d) = \alpha_{i5} \mathbf{1}\{d \in \mathcal{J}_{i5}\} \sigma_{id} s_{id}^{\text{ev}} \sum_{t \in \mathcal{E}_{i5\ell}} \kappa_{i5}(u - t). \quad (29)$$

The shared structure fixes the affected targets, event signs, pulse family, width, and jitter fraction. The level coordinate is the mean-gap fraction of the maximum legal gap; higher levels are sparser. A deterministic compatible phase retains at least three history events and an event on an observed future position. The common gain α_{i5} is fixed across levels and satisfies the history-distance and minimum future-effect gates.

B.7 CD6: Common Factor

Standardize the target history as $Z_i(u, d) = (H_i[u, d] - \mu_{id})/\sigma_{id}$ and write its singular value decomposition as $Z_i = U_i \Sigma_i V_i^\top$. Candidate loading vectors are masked and renormalized members of

$$\mathcal{B}_i = \{v_1, v_1 \pm 0.55v_2, v_1 \pm 0.45v_3\},$$

$$\mathcal{W}_i = \{\text{mask}_\eta(w) : w \in \mathcal{B}_i, \eta \in \{0.10, 0.20, 0.35\}\}. \quad (30)$$

where v_j is row j of $V_i^\top$ and mask_η removes loadings below fraction η of the largest absolute loading while retaining at least three targets. For a selected loading w_{id} , the history-fitted factor $Z_i w_{id}$ supplies a stable harmonic carrier $f_i(u) = a_i^f \sin(\omega_i^f u) + b_i^f \cos(\omega_i^f u)$.

The controlled component is

$$G_{i6}^X(u, d) = \mathbf{1}\{d \in \mathcal{J}_{i6}\} \sigma_{id} w_{id} f_i(u), \quad \Delta X_{i6\ell} = \frac{\lambda_{i6\ell}}{q_{i6}} G_{i6}^X. \quad (31)$$

The seed selects one qualified loading-carrier pair from the bounded leading pool; the loading, participating targets, and carrier remain fixed across levels. The level coordinate is normalized common-factor magnitude. This construction controls observable co-movement and does not assert a causal latent factor.

B.8 CD7: Cross-Series Dependence

For each ordered driver p , responder d , and positive lag τ , CD7 fits the history-only predictive comparison

$$z_{i,d}(u) = \beta_0 + \beta_1 z_{i,d}(u - 1) + \beta_2 z_{i,p}(u - \tau) + \epsilon_u,$$

$$G_{i7}^X(u, j) = \mathbf{1}\{j = d\} \sigma_{id} \beta_2 \tilde{z}_{i,p}(u - \tau), \quad (32)$$

$$\Delta X_{i7\ell} = \frac{\lambda_{i7\ell}}{q_{i7}} G_{i7}^X.$$

Here $z_{i,d}$ is the standardized target history, and $\tilde{z}_{i,p}$ equals the observed driver history before the forecast origin and its history-fitted linear continuation afterward. Candidate edges must improve predictive fit over the responder-only model, then pass source-distance feasibility; the seed selects one retained edge from the bounded leading pool. The driver, responder, lag, and coefficient β_2 are fixed across levels, while $\lambda_{i7\ell}$ controls transfer magnitude. The edge is predictive rather than causal.

B.9 CD8: Covariate Response

CD8 selects a legal dynamic covariate q , target d , event set $\mathcal{E}_{i8}$, covariate sign s_i^U , response sign s_i^X , terminal pre-origin impulse t_0 with multiplier γ_0 , and a one-sided response kernel. With delay δ and half-life parameter h , the peak-normalized kernel is

$$\bar{k}_{i8}(r) = \begin{cases} 0, & r < \delta, \\ \exp[-(\log 2)(r - \delta)/h], & \text{exponential}, \\ \frac{r - \delta}{h} \exp[1 - (r - \delta)/h], & \text{delayed gamma}, \\ \max\left(0, 1 - \frac{r - \delta}{2h}\right), & \text{triangular}, \end{cases} \quad (33)$$

The peak-normalized kernel is

$$k_{i8}(r) = \frac{\bar{k}_{i8}(r)}{\max_{v \geq 0} \bar{k}_{i8}(v)}, \quad \max_{r \geq 0} k_{i8}(r) = 1. \quad (34)$$

Define the signed impulse path

$$I_{i8}(u) = s_i^U \left[\sum_{t \in \mathcal{E}_{i8}} \mathbf{1}\{u = t\} + \gamma_0 \mathbf{1}\{u = t_0\} \right]. \quad (35)$$

Let σ_{iq}^U be the source covariate scale. The raw covariate and target components and their shared level scaling are

$$G_{i8}^U(u, j) = \mathbf{1}\{j = q\} \sigma_{iq}^U I_{i8}(u),$$

$$G_{i8}^X(u, j) = \mathbf{1}\{j = d\} \sigma_{id} s_i^X \sum_{v \leq u} I_{i8}(v) k_{i8}(u - v), \quad (36)$$

$$(\Delta X_{i8\ell}, \Delta U_{i8\ell}) = \frac{\lambda_{i8\ell}}{q_{i8}} (G_{i8}^X, G_{i8}^U).$$

Repeated historical impulses and the terminal impulse use the same kernel. A future covariate impulse is included only when the task declares that covariate known over the horizon; past-only covariates stop at the forecast origin. The seed selects one qualified descriptor from the bounded candidate pool, and $\lambda_{i8\ell}$ jointly scales the covariate impulse and target response without changing their signs, timing, target, covariate, or kernel.

B.10 Connection to the Evaluator

For every capability dimension and level, the Reference–diagnostic-probe pair supplies the truth and forecast changes ΔY_{ik} and $\Delta \hat{Y}_{ik}$ used by paired forecast-change NRMSE in Equation (12). Reference MASE, diagnostic-probe MASE, paired forecast-change NRMSE, and Input Sensitivity (Δ MASE) remain four separate result objects. The CD6–CD8 Input Sensitivity comparison uses an independent full-versus-ablated inference pair whose assessed target is the highest-absolute-loading target for CD6, the responder for CD7, and the paired target for CD8. Positive values mean that auxiliary-input removal increases assessed-target error. This result describes the executed input path and does not by itself identify a causal or internal model mechanism.

C Experimental Scope and Supporting Results

This appendix records the support and aggregation contracts behind the four experimental blocks in the main paper and provides the compact numerical evidence omitted from the main figures. All values use the reported public model configurations only; unsupported model–suite views remain nonnumeric rather than being imputed.

C.1 Compute Environment

Experiments used a standard KVM-based compute-node configuration with 48 Intel Xeon Gold 6530 vCPUs, 188 GiB of system memory, and four NVIDIA GeForce RTX 5090 GPUs with 32,607 MiB of memory per GPU, running Ubuntu 24.04.4 LTS. The CAFE data-generation, validation, and aggregation pipeline used Python 3.12.3. Model inference services used Python 3.11.15, PyTorch 2.10.0, CUDA 12.8, cuDNN 9.10, and NVIDIA driver 575.57.08. This profile describes the standard per-node hardware and software environment; it does not specify request-level allocation or close the incomplete checkpoint, service-revision, numerical-precision, pre-processing, determinism, and replay identities for every reported result.

C.2 Main-Evaluation Support and Aggregation

The main evaluation uses one frozen augmentation realization for each suite. Table 1 gives the horizon and model support used in the reported evaluation. Every admitted capability contains five ordered levels. TiREx-2 is reported only on GIFT-Short and FEV-Mini20 because the medium- and long-range horizons exceed its supported output length; the other five configurations are present in all four suites. Timer-3.5 denotes the evaluated Timer-S1 checkpoint [2].

The three reported metrics are lower-is-better but have distinct populations. Reference MASE is the equal mean of task-level MASE over admitted References. For diagnostic-probe MASE,

Table 1: Support of the current four-suite evaluation. The public roster is Chronos-2, Timer-3.5, TimesFM 2.5, TiREx-2, Moirai 2, and Toto 2.0.

Suite	Horizon	Caps.	Models
GIFT-Short	30–60	8	All six
GIFT-Medium	480–600	8	Five; no TiREx-2
GIFT-Long	720–900	8	Five; no TiREx-2
FEV-Mini20	5–288	8	All six

each capability–level cell is the equal mean of eligible task-level probe MASE; its descriptive macro then gives the five levels equal weight within each capability and the eight capabilities equal weight. Paired forecast-change NRMSE first pools standardized squared forecast-change and truth-change terms within each eligible task as defined in Equation (12); capability–level cells then average the task-level ratios equally, followed by the same level- and capability-equal display macro. NRMSE = 1 is the exact zero-response reference. The two macros summarize the diagnostic grid but do not replace its capability–level cells or define a benchmark-independent leaderboard.

C.3 Expanded Six-Model Ten-Seed Stability

The stability study holds fixed a GIFT-Short panel, eight capabilities, five levels, and six model configurations while refreshing the augmentation seed ten times. For each model and seed, the displayed score is the equal mean over the 40 capability–level cells. Table 2 reports the resulting seed mean, sample standard deviation, rank behavior, and first-place count.

For the six-model analysis reported here, let $m = 10$ denote the regenerated test sets and $n = 6$ the model configurations. Within each test set s , the models are ranked by their capability- and level-equal aggregate NRMSE, with rank 1 assigned to the lowest score. Let r_{sj} be the rank of model j , R_j its rank sum across test sets, and $\bar{R}$ the mean rank sum. Kendall’s coefficient of concordance is

$$R_j = \sum_{s=1}^m r_{sj}, \quad \bar{R} = \frac{m(n+1)}{2}, \quad (37)$$

$$W = \frac{12 \sum_{j=1}^n (R_j - \bar{R})^2}{m^2(n^3 - n)}.$$

Here, $W \in [0, 1]$, with larger values indicating stronger agreement among the test-set rankings and $W = 1$ denoting complete rank agreement. No exact score ties occur in the observed six-model bank, so Equation (37) requires no tie correction. Because W depends only on ranks, it summarizes aggregate ordering rather than score invariance, capability–level winner invariance, or reliability on unseen regenerations. The five-model common-support result in the main paper applies the same calculation with $n = 5$; its $W = 0.940$ is therefore distinct from the six-model value reported below.

Across the ten complete six-model rankings, W is 0.921; the 45 pairwise seed-ranking comparisons have mean Spearman correlation 0.912 and minimum 0.771. Twelve of 40 capability–level cells retain the same winner in every realization, and the modal winner occupies 79.3% of seeds on average. Thus, the aggregate leader and

Table 2: Paired forecast-change NRMSE stability over ten augmentation seeds. Lower is better; the bracket gives the observed rank range.

Model	Mean $\pm$ SD	Mean rank [range]	Top-1 / 10
Chronos-2	0.529 $\pm$ 0.020	1.0 [1–1]	10
TimesFM 2.5	0.563 $\pm$ 0.012	2.3 [2–3]	0
Timer-3.5	0.568 $\pm$ 0.011	2.7 [2–3]	0
TiREx-2	0.630 $\pm$ 0.022	4.4 [4–6]	0
Moirai 2	0.638 $\pm$ 0.009	4.8 [4–6]	0
Toto 2.0	0.671 $\pm$ 0.024	5.8 [5–6]	0

broad ranking structure are stable even though close models and local cells retain useful realization-level variation.

C.4 Input Sensitivity via Direct Auxiliary-Input Removal

For each eligible full/removed pair, we report Input Sensitivity as $\Delta\text{MASE} = \text{MASE}_{\text{removed}} - \text{MASE}_{\text{full}}$ on the assessed target. Positive values mean that the auxiliary input reduced forecast error. The analysis averages five levels within each task and then averages tasks equally; uncertainty is a descriptive task-bootstrap 95% interval. All expected pairs are present. Table 3 lists all six of the 27 evaluated suite-model-capability cells whose intervals exclude zero.

The last row is directionally positive on seven of eight tasks, but its task-equal mean is amplified by strong task heterogeneity: the corresponding task median is 0.014. Reporting both the full heatmap and these intervals is therefore important for separating consistent small benefits from heterogeneous large effects.

C.5 Single-Realization Fine-Tuning Details

The fine-tuning study uses CAFE-generated GIFT-Short probes and reports one training realization throughout. We compare two complete Chronos-2 adaptation recipes: the point-forecast recipe (standard quantile training on individual probes) and the paired-change recipe (Reference-diagnostic-probe training that optimizes forecast-change NRMSE). They also differ in data representation, learning rate, and effective training schedule; consequently, this is a recipe comparison, not a loss-only ablation. The reported curves use the frozen macro-stratum aggregation over dataset, capability, and level strata and are not numerically interchangeable with the task-equal main-evaluation macros.

At 40k, the point-forecast recipe improves MASE by 5.82% but leaves NRMSE 2.88% above baseline after an earlier improvement. The paired-change recipe lowers final NRMSE by 25.66% while raising MASE by 21.15%. The complete trajectories therefore exhibit opposing metric changes under these two single-seed recipes and motivate matched studies of adaptation objectives that preserve both absolute accuracy and response fidelity.

C.6 Complete Numerical Data for the Reported Analyses

This section provides the numerical values underlying the capability, stability, auxiliary-input, and fine-tuning displays in the main paper. The values retain the populations and aggregation rules declared above: they are descriptive outputs of the recorded analysis

package, not a new confirmatory analysis or an expansion of the evaluated support.

C.7 Complete Capability-Level Main-Evaluation Data

Tables 5–8 report the point values underlying the capability heatmaps in the main paper. Each cell gives diagnostic-probe MASE / paired forecast-change NRMSE (P/N). Within a capability-level cell, tasks receive equal weight; the displayed capability value then averages the five levels equally. N is the number of contributing tasks. Boldface marks the lowest value within a suite-capability-metric comparison, and a dash denotes an unsupported model-suite view.

Table 3: Direct removal cells with descriptive task-bootstrap 95% intervals excluding zero. Positive Δ MASE indicates that the full auxiliary input helped; negative values indicate a net increase in error from exposing it.

Suite	Capability	Model	Tasks	Δ MASE	95% interval
GIFT-Short	Common Factor	Chronos-2	4	0.038	[0.009, 0.068]
GIFT-Short	Common Factor	TiREx-2	4	-0.004	[-0.006, -0.001]
GIFT-Long	Covariate Response	Chronos-2	3	-0.005	[-0.010, -0.002]
FEV-Mini20	Covariate Response	TimesFM 2.5	6	0.037	[0.004, 0.079]
FEV-Mini20	Cross-Series Dependence	Chronos-2	8	0.013	[0.004, 0.024]
FEV-Mini20	Cross-Series Dependence	Toto 2.0	8	0.505	[0.002, 1.495]

Table 4: Single-realization Chronos-2 checkpoints for the point-forecast and paired-change recipes. “Best” is the lowest observed value including step 0; lower is better.

Recipe	MASE at 0	Best MASE (step)	MASE at 40k	NRMSE at 0	Best NRMSE (step)	NRMSE at 40k
Point forecast	1.020	0.961 (40k)	0.961	0.464	0.431 (16k)	0.477
Paired change	1.020	1.020 (0)	1.236	0.464	0.344 (36k)	0.345

Table 5: Capability-level P/N values on GIFT-Short. Lower is better for both metrics; NRMSE = 1 is the zero-response reference.

CD	N	Chronos-2	Timer-3.5	TimesFM 2.5	TiREx-2	Moirai 2	Toto 2.0
CD1	18	2.132 / 0.082	1.966 / 0.243	1.898 / 0.216	1.843 / 0.256	2.152 / 0.215	1.876 / 0.243
CD2	20	2.370 / 0.779	2.758 / 0.946	2.499 / 0.802	2.532 / 0.844	4.171 / 1.010	1.934 / 0.637
CD3	19	1.526 / 0.529	1.738 / 0.674	1.462 / 0.679	1.484 / 0.580	1.853 / 0.740	1.572 / 0.571
CD4	20	1.703 / 0.096	1.717 / 0.133	1.627 / 0.091	1.597 / 0.071	1.910 / 0.134	1.574 / 0.144
CD5	20	2.929 / 0.905	2.872 / 0.914	2.833 / 0.837	2.745 / 0.972	4.211 / 0.965	2.699 / 0.955
CD6	4	0.618 / 0.257	0.644 / 0.301	0.648 / 0.333	0.658 / 0.259	0.695 / 0.353	0.625 / 0.291
CD7	8	2.895 / 0.989	2.982 / 1.043	2.896 / 1.187	2.865 / 1.005	3.060 / 1.044	2.746 / 1.460
CD8	6	3.638 / 0.626	3.833 / 0.554	3.491 / 0.495	3.485 / 0.850	5.907 / 0.684	3.522 / 0.632

Table 6: Capability-level P/N values on GIFT-Medium. Lower is better for both metrics; NRMSE = 1 is the zero-response reference.

CD	N	Chronos-2	Timer-3.5	TimesFM 2.5	TiREx-2	Moirai 2	Toto 2.0
CD1	9	2.982 / 0.051	2.808 / 0.264	2.880 / 0.259	–	3.469 / 0.293	12.785 / 0.590
CD2	11	5.309 / 1.102	5.533 / 1.206	4.016 / 0.961	–	10.671 / 1.729	10.488 / 1.131
CD3	10	7.051 / 0.787	7.292 / 1.130	3.507 / 0.980	–	10.616 / 1.565	9.955 / 1.296
CD4	11	2.378 / 0.053	2.581 / 0.191	2.453 / 0.131	–	4.026 / 0.276	7.597 / 0.176
CD5	11	2.851 / 1.402	3.039 / 1.738	2.936 / 1.322	–	3.592 / 1.115	6.459 / 1.556
CD6	4	0.919 / 0.257	0.952 / 0.474	0.966 / 0.524	–	1.324 / 0.571	1.163 / 0.377
CD7	6	7.690 / 0.992	8.041 / 1.070	8.089 / 1.053	–	9.902 / 1.051	15.988 / 1.189
CD8	4	5.151 / 0.533	6.848 / 0.498	6.849 / 0.373	–	9.773 / 0.457	19.072 / 0.550

Table 7: Capability-level P/N values on GIFT-Long. Lower is better for both metrics; NRMSE = 1 is the zero-response reference.

CD	N	Chronos-2	Timer-3.5	TimesFM 2.5	TiREx-2	Moirai 2	Toto 2.0
CD1	10	3.597 / 0.062	3.549 / 0.191	3.682 / 0.124	–	4.609 / 0.213	16.115 / 0.257
CD2	11	6.252 / 0.914	6.780 / 1.168	4.890 / 0.966	–	9.542 / 1.638	21.869 / 0.998
CD3	10	4.860 / 0.859	8.021 / 1.223	5.632 / 0.961	–	9.717 / 1.190	23.432 / 1.119
CD4	11	2.934 / 0.064	3.100 / 0.175	3.253 / 0.106	–	4.442 / 0.326	16.994 / 0.265
CD5	11	3.131 / 1.051	3.087 / 1.118	3.185 / 1.069	–	4.636 / 1.683	16.407 / 1.184
CD6	4	1.039 / 0.385	1.144 / 0.739	1.113 / 0.772	–	1.749 / 1.084	1.324 / 0.663
CD7	6	6.669 / 0.893	7.005 / 0.995	7.446 / 1.058	–	10.128 / 1.019	36.761 / 1.326
CD8	3	9.909 / 0.397	13.459 / 0.594	9.512 / 0.335	–	14.300 / 0.490	53.238 / 0.857

Table 8: Capability-level P/N values on FEV-Mini20. Lower is better for both metrics; NRMSE = 1 is the zero-response reference.

CD	N	Chronos-2	Timer-3.5	TimesFM 2.5	TiREx-2	Moirai 2	Toto 2.0
CD1	18	1.938 / 0.176	1.807 / 0.181	1.469 / 0.111	1.736 / 0.130	1.625 / 0.087	1.591 / 0.124
CD2	20	2.656 / 1.202	3.003 / 1.143	2.582 / 0.954	2.703 / 1.042	3.144 / 1.213	1.839 / 0.809
CD3	20	2.190 / 0.812	2.140 / 0.756	1.744 / 0.734	2.040 / 0.794	2.045 / 0.797	1.659 / 0.663
CD4	20	1.913 / 0.213	1.917 / 0.163	1.562 / 0.120	1.853 / 0.136	1.647 / 0.103	1.603 / 0.110
CD5	18	2.188 / 0.934	1.987 / 0.963	1.712 / 0.864	2.154 / 0.894	2.055 / 1.062	1.731 / 0.852
CD6	9	1.766 / 0.286	2.044 / 0.359	1.630 / 0.301	1.819 / 0.291	1.791 / 0.380	1.663 / 0.295
CD7	9	1.766 / 0.718	2.100 / 0.782	1.678 / 0.763	1.693 / 0.717	1.711 / 0.834	1.826 / 0.752
CD8	8	3.073 / 0.776	3.046 / 0.642	2.480 / 0.691	2.972 / 0.651	3.025 / 0.674	2.303 / 0.551

Table 9: Spearman rank correlations among Reference MASE (R), probe MASE (P), and NRMSE (N) for the displayed model sets; intervals are not estimated.

Suite	Models	$\rho_{R,P}$	$\rho_{R,N}$	$\rho_{P,N}$
GIFT-Short	6	0.943	-0.086	0.029
GIFT-Medium	5	0.700	0.800	0.800
GIFT-Long	5	0.900	0.800	0.900
FEV-Mini20	6	0.657	0.543	0.771

Table 10: Local-winner stability across ten GIFT-Short seeds. Unanimous levels retain one lowest-NRMSE model in all seeds; modal fractions are descriptive.

CD	Unanimous / 5	Mean modal	Min modal	Level-modal models
CD1	5	1.00	1.00	Chronos-2 (5)
CD2	0	0.82	0.70	TimesFM 2.5 (1), Toto 2.0 (4)
CD3	5	1.00	1.00	Chronos-2 (5)
CD4	0	0.68	0.50	TiREx-2 (3), TimesFM 2.5 (1), Chronos-2 (1)
CD5	0	0.74	0.50	TimesFM 2.5 (4), Chronos-2 (1)
CD6	0	0.64	0.50	Chronos-2 (5)
CD7	0	0.60	0.50	TiREx-2 (4), Chronos-2 (1)
CD8	2	0.86	0.70	TimesFM 2.5 (5)

Table 11: Complete Input Sensitivity results for the 27 evaluated suite–model–capability cells. Each estimate is $\text{MASE}_{\text{ablated}} - \text{MASE}_{\text{full}}$ with a descriptive task-bootstrap 95% interval; positive values favor the full auxiliary input.

Suite	CD	Model	Tasks	ΔMASE [95% interval]	Task median	Positive tasks
GIFT-Short	CD6	Chronos-2	4	0.038 [0.009, 0.068]	0.042	3/4
GIFT-Short	CD6	TiREx-2	4	-0.004 [-0.006, -0.001]	-0.005	1/4
GIFT-Short	CD6	Toto 2.0	4	0.005 [-0.012, 0.030]	-0.004	2/4
GIFT-Short	CD7	Chronos-2	8	0.012 [-0.001, 0.030]	0.002	6/8
GIFT-Short	CD7	TiREx-2	8	0.003 [-0.002, 0.008]	0.001	5/8
GIFT-Short	CD7	Toto 2.0	8	0.031 [-0.002, 0.077]	-0.001	4/8
GIFT-Short	CD8	Chronos-2	6	-0.001 [-0.012, 0.010]	0.001	4/6
GIFT-Short	CD8	TiREx-2	6	0.001 [-1.21 $\times 10^{-4}$, 0.003]	2.29 $\times 10^{-4}$	4/6
GIFT-Medium	CD6	Chronos-2	4	-0.007 [-0.082, 0.050]	0.012	3/4
GIFT-Medium	CD6	Toto 2.0	4	-0.024 [-0.095, 0.038]	-0.014	2/4
GIFT-Medium	CD7	Chronos-2	6	0.005 [-0.048, 0.048]	0.011	4/6
GIFT-Medium	CD7	Toto 2.0	6	-0.039 [-0.093, 0.002]	-0.016	1/6
GIFT-Medium	CD8	Chronos-2	4	0.011 [-0.002, 0.033]	0.001	2/4
GIFT-Long	CD6	Chronos-2	4	0.034 [-0.064, 0.153]	0.011	3/4
GIFT-Long	CD6	Toto 2.0	4	-0.012 [-0.029, 0.005]	-0.012	1/4
GIFT-Long	CD7	Chronos-2	6	-0.006 [-0.092, 0.082]	-0.014	2/6
GIFT-Long	CD7	Toto 2.0	6	0.195 [-0.034, 0.629]	-0.008	3/6
GIFT-Long	CD8	Chronos-2	3	-0.005 [-0.010, -0.002]	-0.003	0/3
FEV-Mini20	CD6	Chronos-2	8	0.048 [-0.005, 0.128]	0.003	6/8
FEV-Mini20	CD6	TiREx-2	8	-0.020 [-0.056, 0.001]	-0.003	2/8
FEV-Mini20	CD6	Toto 2.0	8	0.135 [-0.010, 0.402]	-0.001	4/8
FEV-Mini20	CD7	Chronos-2	8	0.013 [0.004, 0.024]	0.009	7/8
FEV-Mini20	CD7	TiREx-2	8	-0.010 [-0.029, 0.003]	0.001	5/8
FEV-Mini20	CD7	Toto 2.0	8	0.505 [0.002, 1.495]	0.014	7/8
FEV-Mini20	CD8	Chronos-2	8	2.935 [-0.299, 8.982]	0.150	5/8
FEV-Mini20	CD8	TimesFM 2.5	6	0.037 [0.004, 0.079]	0.028	5/6
FEV-Mini20	CD8	TiREx-2	8	0.038 [-0.530, 0.706]	0.004	5/8

Table 12: Complete single-realization Chronos-2 trajectories for the point-forecast and paired-change recipes. Values use the frozen macro-stratum aggregation on training seed 2026082701; lower is better, and no repeated-training-seed interval is available.

Step	Point-forecast MASE	Point-forecast NRMSE	Paired-change MASE	Paired-change NRMSE
0	1.020	0.464	1.020	0.464
4k	0.998	0.441	1.189	0.375
8k	0.991	0.449	1.175	0.354
12k	0.982	0.434	1.185	0.349
16k	0.990	0.431	1.218	0.348
20k	0.981	0.444	1.214	0.350
24k	0.976	0.449	1.233	0.347
28k	0.970	0.462	1.239	0.349
32k	0.970	0.472	1.231	0.344
36k	0.965	0.480	1.232	0.344
40k	0.961	0.477	1.236	0.345

D Evidence, Experiment Identity, and Coverage

This appendix binds the paper-facing results to the identities and compact records that were available for the current supplement build. It distinguishes the reviewer-facing packaging revision from the source revisions that produced individual experiment blocks. The packaged source snapshot was inspected at commit 55bd36e3aae9ce523d97faabde6e1243969d42da on September 2, 2026. This snapshot contains reproduction scripts, configurations, compact snapshots, tables, figures, and the interactive result browser; it does not retroactively replace the upstream source revision stored by each recorded run.

D.1 Recorded Experiment Identities

Table 13 records the experiment IDs and source identities exposed by the reviewer-facing configuration. The three GIFT main ranges share one recorded augmentation seed and source revision, whereas FEV, the ten-seed stability bank, and fine-tuning have distinct source and control identities. These blocks therefore remain separate evidence populations.

The fine-tuning environment is separately frozen because it extends the upstream Chronos training repository. Its recorded environment used Python 3.12.3, PyTorch 2.11.0 with CUDA 12.8, float32 checkpoint evaluation, a training context of 2,048, and an evaluation context of 8,192. These identities differ from the PyTorch 2.10.0 inference-service environment reported for the main benchmark runs and should not be merged into one runtime claim.

D.2 Claim-to-Artifact Map

Table 14 maps each claim family to the compact reviewer-facing records and the current validation boundary. The local paper-facing derivation package is rooted at `analysis/experiment_rewrite_20260902`; the public reconstruction package is rooted at `paper_results/work`. Basenames are shown to keep the table readable.

D.3 Execution Inventory and Analysis Coverage

Table 15 reproduces the current public experiment inventory. “References” and “probes” count recorded evaluation objects, whereas “predictions” counts public-model prediction records. Unsupported medium- and long-range TiREx-2 views were not issued and are not counted as failures. The zero failure counts apply to issued requests in this inventory; they do not erase metric-specific nonnumeric states such as low-truth-effect NRMSE or a capability that is inapplicable to a task.

The compact supplement derivation contains 22 suite-model rows, 176 capability rows, and 880 capability-level rows for the main block. The stability block contains 60 seed-model rows, 45 pairwise seed-rank comparisons, and 40 capability-level winner-consistency rows. The input-removal block contains 27 suite-model-capability summaries, 825 task-level rows, and 165 coverage rows, with zero missing expected paper-facing pairs. The fine-tuning block contains 44 curve rows, 32 final capability changes, 16 condensed key-checkpoint rows, and two recipe summaries. These are downstream record counts, not independent experimental units; task is the equal-weight unit in the main and removal

analyses, seed indexes the observed stability panel, and the fine-tuning curves share one training realization.

Unavailable, unsupported, not run, failed, invalid, and metric-undefined states remain nonnumeric. They are not filled with zero and do not silently change a comparison denominator. NRMSE additionally distinguishes scoreable instances from low-truth-effect and no-observed-effect states as defined in Appendix A. Table 16 makes this partition explicit for the main block. An attempted group is one official instance paired with one of CD1-CD8; a qualified group contributes exactly five treatment contracts. “Excluded” is the complement of qualified groups in this attempted-group denominator. The NRMSE columns partition every generated treatment, so the residual analysis-invalid count is zero.

The two public machine-readable ledgers are `experiment_inventory.csv`, keyed by suite, and `capability_availability.csv`, keyed by suite and capability; the source manifests retain task and instance identifiers. They reconcile to 21,649,666 completed public-model prediction records and zero failed issued requests. TiREx-2 was unsupported for the medium- and long-range suites, so two suite-model cells (129,884 baseline, treatment, and removal views) were not issued rather than counted as failures. For stability, `suite_coverage_audit.csv` records the same ten task identifiers for every seed and 240 public capability-level-model cells per seed; all 60 seed-model macro rows are present. For input removal, the public ledger contains 165 eligible suite-dataset-model-capability identifiers, 825 task-level rows, 27 summaries, and zero missing expected pairs. For fine-tuning, the fit and evaluation manifests contain 50,535 and 48,365 eligible treatments; the reported realization contains all 44 objective-checkpoint-metric curve rows, 32 final capability changes, and 16 condensed key-checkpoint rows. Unsupported and qualification states are not applicable to the latter three downstream ledgers because they consume already qualified main-run contracts; missing expected identifiers, failed execution, and nonnumeric reported cells are each zero in the frozen records.

D.4 Completed Checks

The current local derivation audit reports a maximum absolute capability reconciliation error of 6.1×10^{-10} and a maximum macro reconciliation error below 7.6×10^{-11} for the main tables. Stability macro reconciliation is exact at the recorded precision, and all expected input-removal pairs are present. These checks support the displayed downstream numbers. They do not replay generation, model inference, or training.

On September 2, 2026, the `verify` subcommand at commit 55bd36e reported that all 13 frozen paper artifacts passed; the release-candidate package builder also validated all 14,106 payload files against its internal checksum list. These checks establish the integrity of the frozen records and their paper-facing derivations without relabeling them as end-to-end or independent reproduction.

E Artifact and Reproduction Guide

This appendix is the reviewer-facing map from the public source snapshot to verification, analysis reconstruction, and end-to-end

Table 13: Recorded identities of the experiment blocks used by the current paper-facing package. A packaging revision is not substituted for the source revision bound by a run.

Block	Recorded experiment identity	Deterministic control	Bound source identity
GIFT main	gift-v15-short-qualified-feasible-seed2026082701; analogous medium and long IDs	Augmentation seed 2026082701; 20 short tasks and 11 tasks in each longer range	CAFE 0e8590ee466046941656ede43dd1edc61a92260a
FEV main	fev-mini20-full-v6	Augmentation seed 2026082601; FEV Mini20	CAFE bd1e42d943182b1ab5bf369f415f0fa0f72dd776; FEV 0.8.0 at f1afffbf97bc51a4a233080d331633c6f7ab32f6; data f71c0fff4cf81283a2c43e7f3a73aa4f9826aef8
Input removal	gift-v15-seed2026082701-target-only-v1; fev-mini20-full-v6-target-only-v1	Same treatment contracts as the corresponding main run; assessed target history and future fixed	Sources are the bound GIFT-Short and FEV main experiments above
Stability	gift-v15-short-stability10-head78ef32f-seed{seed}	Fixed ten-task GIFT-Short panel; seeds 2026082701–2026082710	CAFE 78ef32fe1dd71583b9312a73f17e4cb81126917f
Fine-tuning	Fit and evaluation contract trees ...seed2026082701-r1 and ...seed2026082702-r1	Training seed 2026082701; two complete recipes; 40,000 steps; checkpoints every 4,000 steps	Chronos 8589d1988e9676817548e9626738ff06b6ca6370; base model revision 29ec3766d36d6f73f0696f85560a422f50e8498c

Table 14: Claim-to-artifact map for the current supplement. “Regenerated” means that the stated downstream analysis-to-display relation was checked; it does not mean that model execution was replayed.

Claim family	Compact numerical records	Reviewer-facing reconstruction	Current evidence boundary
CD1–CD8 constructions	Section B; generation contracts; mechanisms.py	Generation, validation, and replay commands in the public source snapshot	Implemented construction contract; no model-internal mechanism identification
Reference/probe/response ranking differences	main_suite_model_metrics.csv; main_metric_rank_correlations.csv	official_mase_suite.csv; effect_nrmse_derived_macro.csv	Downstream tables regenerated; current aggregation is CAFE-native task-equal, not source-leaderboard parity
Capability-resolved profiles	main_capability_metrics.csv; main_capability_level_metrics.csv	treatment_mase_by_suite_model_capability_level.csv; capability heatmap	Capability and macro reconciliation checked; capability-level cells remain primary
Ten-seed stability	stability_seed_model_macro.csv; stability_seed_rank_similarity.csv; winner consistency table	model_overall_stability_extended.csv; stability figure	Descriptive fixed-panel seed bank; no claim about unseen-seed reliability
Auxiliary-input removal	ablation_target_only_summary.csv; task-level and coverage tables	ablation_collapsed_summary_public.csv; ablation figure	All expected paper-facing pairs present; association under the declared removal, not causal attribution to an internal mechanism
Fine-tuning trade-off	Curves, key checkpoints, and final capability changes for training seed 2026082701	key_checkpoints.csv; absolute curves; frozen overlay	One training realization and two unequal complete recipes; not a loss-only ablation

Table 15: Current main-experiment inventory from the frozen reviewer-facing table.

Suite	Tasks	References	Probes	Models	Predictions	Failures	Horizon
GIFT-Short	20	120,784	2,662,855	6	16,938,804	0	30–60
GIFT-Medium	11	4,077	70,995	5	380,810	0	480–600
GIFT-Long	11	2,902	50,090	5	268,610	0	720–900
FEV-Mini20	20	23,062	563,570	6	4,061,442	0	5–288

Table 16: Main-block generation and NRMSE coverage ledger. Counts are treatment groups through “excluded” and treatment contracts thereafter. The attempted-group denominator is official instances $\times 8$; each qualified group contributes five treatments.

Suite	Attempted groups	Qualified groups	Excluded groups	Generated treatments	Scoreable	Low truth effect	No observed effect	Invalid
GIFT-Short	966,272	532,571	433,701	2,662,855	2,660,523	2,332	0	0
GIFT-Medium	32,616	14,199	18,417	70,995	70,995	0	0	0
GIFT-Long	23,216	10,018	13,198	50,090	50,090	0	0	0
FEV-Mini20	184,496	112,714	71,782	563,570	555,816	414	7,340	0

execution. The three levels intentionally have different prerequisites and evidence meanings. Verifying frozen files checks identity and shape, reconstruction reruns the paper-facing derivations from compact snapshots, and end-to-end execution regenerates treatments and forecasts. None of these levels is called completed merely because its command is documented.

E.1 Current Public Entry Point and Package Scope

The current public entry point is <https://github.com/thulab/CAFE/tree/research>; the release-candidate source snapshot inspected for this supplement is commit 55bd36e3aae (the full identity appears in Appendix D). The repository is Apache-2.0 licensed and contains the CAFE source, dependency lock, machine-readable paper configurations, compact analysis snapshots, expected-output manifest, reconstruction scripts, the interactive result browser, and the

frozen Chronos-2 fine-tuning overlay. The public projection excludes the unreleased internal model and does not use it in the paper-facing tables, ranks, or figures.

The complete external reviewer artifact is specified as one approximately 20 GB archive containing the exact source snapshot, authentic GIFT-Eval and FEV source snapshots, main experiment contracts and predictions, all ten stability trees, fine-tuning contract trees and LoRA checkpoints, and target-only summaries. Its top level is required to contain README.md, MANIFEST.json, SHA256SUMS, the source snapshot, data, experiment trees, stability summary, and fine-tuning outputs. The designated distribution folder is <https://cloud.tsinghua.edu.cn/d/77e1d26573a347e89b6b/>. The local release candidate has the following identity; each multi-line field is the concatenation of its displayed fragments:

```
FILE: cafe-reviewer-
      artifact-v1.zip
BYTES: 22,844,720,139
SCHEMA: cafe_reviewer_
        artifact.v1
SHA-256:
3485293f8ba7b06f
ec17e492b07ca9cb
b80812c3edd5b3f0
d90c1f4cd4c57cf0
CAFE: 55bd36e3aae9ce523d97
      faabde6e1243969d42da
```

The URL is a distribution location; the byte size, digest, schema, and source identity specify the candidate bytes.

E.2 Three Reproduction Levels

The recorded main-run environment used Ubuntu 24.04.4, Python 3.12.3, uv 0.12.0, CUDA 12.8, NVIDIA driver 575.57.08, and RTX 5090 GPUs. The authoritative dependency identity is `uv.lock`; the package versions summarized in Appendix C describe the recorded machine rather than a portability guarantee. Models may be executed sequentially on fewer GPUs if their context and memory requirements are met.

E.3 Verification and Reconstruction Commands

After checking out the frozen release revision, verify the representative output files with:

```
python reproducibility/reproduce.py verify
```

Install the plotting and test dependencies and reconstruct each paper block with:

```
uv sync --extra plots --extra test
python reproducibility/reproduce.py figures main
python reproducibility/reproduce.py figures ablation
python reproducibility/reproduce.py figures stability
python reproducibility/reproduce.py figures finetuning
```

The verification manifest expects four inventory rows, 22 main suite-model MASE rows, 880 capability-level diagnostic-probe MASE rows, six overall stability rows, 40 capability-level winner-consistency rows, 27 input-removal summary rows, and 32 public

fine-tuning key-checkpoint rows, together with the four representative PDF figures. The supplement's 16-row key-checkpoint table is a further condensed display derived from the 32-row public file.

CPU-derived tables are expected to match the committed files exactly in the recorded environment. GPU forecasts and training may vary across CUDA, driver, and kernel versions; acceptance therefore compares prespecified aggregate cells rather than requiring byte-identical weights. Table 18 freezes these gates before reviewer replay. Each comparison joins the candidate and frozen records on their complete identifier key. Coverage counts and non-numeric state categories must match exactly.

A replay fails if a required key is absent, a numeric cell changes to a nonnumeric state, an unsupported or metric-undefined state changes category, a coverage count changes, or any joined cell exceeds its gate. Ranks and figures are regenerated only after cell-level acceptance and are not substitute tolerances for near ties.

E.4 End-to-End Main and Stability Runs

The full pipeline requires local paths for the GIFT-Eval source, FEV Mini snapshot, model weights and runtime, output root, and CUDA devices. The exact main commands are emitted rather than copied into this document:

```
python reproducibility/reproduce.py commands <block>
```

Here, `<block>` is one of `gift_short`, `gift_medium`, `gift_long`, `fev`, or `stability`.

Each emitted main command runs generation, validation, inference, and analysis in that order. The stability command emits ten runs over the fixed task panel using augmentation seeds 2026082701–2026082710. Distributed worker count is a throughput setting; the task panel, augmentation seed, treatment contracts, model configuration, and aggregation must remain fixed.

The bounded GPU smoke test uses `gift_ett1_h`, augmentation seed 2026082701, CD1 only, one official instance, Chronos-2, and device 0. The complete command is:

```
SMOKE=reviewer-smoke-2026082701
OUT="$CAFE_OUTPUT_ROOT"
GIFT="$GIFT_EVAL_DIR"
MODEL="$CAFE_MODEL_ROOT"
CODE="$CAFE_MODEL_CODE_ROOT"
REPRO=reproducibility/reproduce.py
uv run cafe run \
  --experiment-id "$SMOKE" \
  --dataset-id gift_ett1_h --term short \
  --gift-eval-dir "$GIFT" \
  --output-root "$OUT" \
  --augmentation-seed 2026082701 \
  --capabilities trend --models Chronos-2 \
  --backend native --model-root "$MODEL" \
  --model-code-root "$CODE" \
  --devices 0 --max-instances 1 \
  --generation-workers 1 \
  --validation-workers 1 \
  --validation-dataset-workers 1 \
  --preprocess-workers 1 \
  --analysis-workers 1 --disk-budget-gb 5
python "$REPRO" smoke-check \
```

Table 17: Reviewer workflows and declared resource prerequisites.

Level	Declared environment	Output and evidence meaning
Verify frozen artifacts	Linux or macOS; Python 3.11+; CPU; less than 1 GB memory and disk	Checks SHA-256 digests and row counts for representative tables and figures; no analysis or model execution
Reconstruct tables and figures	Linux or macOS; Python 3.11+; uv; 4 CPU cores, 8 GB RAM, and 5 GB free disk recommended	Recomputes four paper-facing analysis blocks from compact snapshots without contacting inference services
End-to-end experiments	Linux x86-64; Python 3.12; CUDA 12.8-compatible NVIDIA driver; 32 GB VRAM per worker; 16 CPU cores, 64 GB RAM, and 100 GB free disk recommended	Regenerates treatments, verifies them, runs public models, and reconstructs analysis; requires source benchmarks, weights, and substantially more time

Table 18: Prespecified numerical acceptance gates for GPU replay. Here x is the frozen reference value; every listed metric is lower-is-better.

Block	Comparison unit	Maximum accepted deviation
Main and stability	Shared suite-model-capability-level cell for probe MASE and NRMSE; suite-model cell for Reference MASE	$\max(10^{-4}, 0.01 x)$
Input removal	Shared suite-model-capability cell for Δ MASE	Absolute deviation 10^{-3}
Checkpoint re-evaluation	Shared objective-checkpoint-metric cell	$\max(5 \times 10^{-4}, 0.01 x)$
Fine-tuning from step zero	Shared objective-checkpoint-metric cell	$\max(2 \times 10^{-3}, 0.02 x)$

"\$OUT/\$SMOKE"

The checker requires one reference instance, five treatment contracts, accepted validation, one completed Chronos-2 model with no failed requests, all four stage contracts, and the compact suite analysis. A successful check reports `smoke check passed` followed by `dataset=gift_ett1_h`, `instances=1`, `treatments=5`, `models=1`, and `stages=4`. Runtime is dominated by the first model load and local download cache; the bounded scope, rather than a wall-clock promise, defines acceptance.

Recovery preserves experiment identity. `experiment.json` fixes the experiment ID, while each stage contract fixes code, configuration, and upstream hashes; any mismatch requires a new experiment ID. Generation, validation, and analysis are immutable once their terminal manifest or report exists. If one of those stages is interrupted before that record is written, only its exact incomplete stage directory may be quarantined and regenerated. Inference resumes with the original command plus `--start-at inference --resume-inference`; it accepts a completed shard only when the recorded hash still matches. Stopped-process `*.tmp` prediction files and `04_analysis/.source_shard_parts/` without an analysis manifest are disposable caches. Completed stages, contracts, prediction shards, and source snapshots are not cleanup targets, and files from distinct experiment roots are never merged.

Fine-tuning preparation reuses a cache only when its local manifest exists; otherwise that exact cache is quarantined and rebuilt from the CaFE contracts. The released runner does not claim optimizer-state recovery, so an interrupted training recipe restarts in a new empty model directory. Completed LoRA checkpoints can be evaluated independently, and incomplete metric parts can be regenerated without repeating training. After recovery, reviewers rerun `smoke-check` or frozen verification and confirm that all stage contracts retain one experiment identity and matching upstream hashes.

E.5 Chronos-2 Fine-Tuning Workflow

The fine-tuning package checks out the pinned Chronos commit, applies the frozen CaFE overlay, verifies the pinned base-model revision, prepares the two recorded contract trees, and then trains

and evaluates both recipes. The complete identities appear in Table 13. The recorded fit and evaluation corpora contain 50,535 and 48,365 eligible treatments, respectively. Both recipes use LoRA, training seed 2026082701, 40,000 optimizer steps, and checkpoints every 4,000 steps; they differ in objective, representation, learning rate, forecast origin, and effective exposure.

Before allocating GPUs, reviewers can inspect the full workflow with the runner's `--dry-run` option. The actual workflow separates prepare, train, and evaluate; reconstruction of the paper-facing curves then uses `python reproducibility/reproduce.py figures finetuning`. The archived LoRA checkpoints permit checkpoint re-evaluation without repeating both training jobs, but this shortcut does not reproduce training.

References

- [1] Rob J. Hyndman and Anne B. Koehler. 2006. Another look at measures of forecast accuracy. *International Journal of Forecasting* 22, 4 (2006), 679–688. <https://doi.org/10.1016/j.ijforecast.2006.03.001>
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